MAT 3103: Computational Statistics and Probability

Chapter 5: Discrete Distribution

Probability Distribution:

A probability distribution represents all possible outcomes of a random variable along with their associated probabilities, providing a structured way to understand uncertainty in real-world scenarios. In research contexts, particularly in Bangladesh, probability distributions are vital for analyzing phenomena such as rainfall patterns, traffic flow, or disease outbreaks. For instance, predicting dengue fever cases based on seasonal data or forecasting river flooding probabilities relies on probability distributions to model and interpret uncertainty effectively. These distributions describe how likely different outcomes are and offer a comprehensive view of the data, which is often more insightful for decision-making and policy development than focusing solely on the most probable outcome.

Discrete probability distribution:

A discrete probability distribution describes the probability of occurrence for each specific value of a discrete random variable. In the context of research in Bangladesh, such distributions play a crucial role in fields like machine learning, natural language processing (NLP), and public health analysis. For instance, discrete probability distributions are essential for modeling binary and multi-class classification problems, such as predicting disease presence (e.g., dengue or tuberculosis) or analyzing literacy rates in various regions. They are also used to evaluate the performance of classification models by calculating confidence intervals. In NLP applications, discrete distributions help model the frequency of words in Bangla text, which can be applied in tasks like sentiment analysis or translation. Moreover, understanding these distributions is critical for selecting activation functions and loss functions in deep learning models, especially for classification tasks tailored to specific local datasets, such as agricultural outcomes or urban traffic patterns.

Binomial distribution:

The binomial distribution is the discrete probability distribution of the number of successes in a sequence of *n* independent yes/no experiments, each of which yields success with probability *p*. Binomial process is a random process in which:

- The process is performed under the same conditions for fixed and finite number of trials.
- Each trial is independent of other trials.
- Each trial has two mutually exclusive outcomes.
- The probability of success, *p*, remains constant from trial to trial (so the probability of failure, *q*, where, $q = 1-p$).

In general, if the random variable *X* follows the binomial distribution with parameters *n* and *p*, we write $X \sim B(n, p)$. The probability of getting exactly *x* successes in *n* trials is given by the probability mass function:

$$P(X = x) = {}^{n}C_{x}\, p^{x}(1-p)^{n-x} \text{ ; } x = 0, 1, 2, \ldots, n.$$

Where, p = the probability of a success.

n = the number of trials.

x = the number of success in n trials.

The binomial distribution is widely used in scenarios where outcomes are binary, such as "success" or "failure." In the context of Bangladesh, it is particularly relevant in fields like quality control, public health, and education. For example, in manufacturing industries, the binomial distribution helps assess the probability of defective versus non-defective items produced on a production line, aiding in maintaining quality standards. In public health, it is used to estimate the likelihood of patient survival or mortality in medical studies, such as analyzing the effectiveness of treatments for diseases like dengue or COVID-19. Similarly, in education, it can help estimate the probability of students passing or failing an examination. These applications highlight the utility of binomial distribution in addressing practical challenges across various sectors.

Note: For binomial distribution: E(X) = np, V(X) = npq. The terms 'success' and 'failure' in probability are merely labels and can often be misleading, as they do not imply positive or negative outcomes. For instance, in the context of manufacturing in Bangladesh, if a machine produces defective items, counting these defects might be labeled as 'successes' if the objective is to identify the number of defective items. Similarly, in public health research, when analyzing the spread of diseases, detecting cases of infection could be considered 'successes,' even though infections are undesirable. These labels depend on the perspective of the analysis and should be interpreted based on the context rather than their conventional meanings.

Problem 1: One prominent physician claim that 70% of those with lung cancer are chain smokers. Ten such patients recently admitted to a hospital. If his assertion is correct, find the probability that

(a) fewer than half are chain smokers.

(b) more than 7 are chain smokers.

Solution:

where:

n = the number of patients,

p = 0.7 is the probability of a patient being a chain smoker.

(a) Probability that fewer than half (i.e., less than 5) of 10 patients are chain smokers

We need to calculate:

$$P(X < 5) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$
$$= \left(\binom{10}{0}(0.7)^0(0.3)^{10-0}\right) + \left(\binom{10}{1}(0.7)^1(0.3)^{10-1}\right) + \left(\binom{10}{2}(0.7)^2(0.3)^{10-2}\right) +$$
$$\left(\binom{10}{3}(0.7)^3(0.3)^{10-3}\right) + \left(\binom{10}{4}(0.7)^4(0.3)^{10-4}\right) = 0.0473$$

(b) Probability that more than 7 of 10 patients are chain smokers

We need to find:

$$P(X > 7) = P(X = 8) + P(X = 9) + P(X = 10)$$
$$= \left(\binom{10}{8}(0.7)^8(0.3)^{10-8}\right) + \left(\binom{10}{9}(0.7)^9(0.3)^{10-9}\right) + \left(\binom{10}{10}(0.7)^{10}(0.3)^{10-10}\right)$$
$$= 0.3828$$

Problem 2: A farmer in Bangladesh plants 25 jute seedlings. Based on past experience, each seedling has a 90% chance of surviving to harvest. What is the probability that:

(a) All 25 seedlings will survive?

(b) At least 20 seedlings will survive?

Solution:

This is another binomial distribution problem with:

$n = 25$ (number of trials, i.e., seedlings)

$p = 0.9$ (probability of success, i.e., seedling survival)

$q = 1 - p = 0.1$ (probability of failure)

(a) $P(X = 25) = \binom{25}{25}(0.9)^{25}(0.1)^{25-25} = 0.0718$

Therefore, the probability that all 25 seedlings will survive is approximately 7.18%.

(b) $P(X \geq 20) = P(X = 20) + P(X = 21) + P(X = 22) + P(X = 23) + P(X = 24) +$

$P(X = 25) = \binom{25}{20}(0.9)^{20}(0.1)^{25-20} + \binom{25}{21}(0.9)^{21}(0.1)^{25-21} + \binom{25}{22}(0.9)^{22}(0.1)^{25-22} +$

$\binom{25}{24}(0.9)^{24}(0.1)^{25-24} + \binom{25}{25}(0.9)^{25}(0.1)^{25-25} = 0.9666$

Therefore, the probability that at least 25 seedlings will survive is approximately 96.66%.

Problem 3: The Bangladesh cricket team is playing a series of 5 one-day matches against India. Based on historical data, Bangladesh has a 40% chance of winning each match. What is the probability that Bangladesh will win:

(a) Exactly 2 matches?

(b) More matches than India?

Solution:

This is a binomial distribution with:

$n = 5$ (number of trials, i. e. , matches)

$p = 0.4$ (probability of success, i. e. , Bangladesh winning)

$q = 1 - p = 0.6$ (probability of failure)

(a) $P(X = 2) = \binom{5}{2}(0.4)^{2}(0.6)^{5-2} = 0.3456$

So, the probability that Bangladesh will win exactly 2 matches is approximately 34.56%.

(b) This means Bangladesh needs to win 3, 4, or 5 matches:

$$P(X > 2) = P(X = 3) + P(X = 4) + P(X = 5)$$

$$= 0.3174$$

So, the probability that Bangladesh will win more matches than India is approximately 31.74%.

Problem 4: A coastal district in Bangladesh has a 60% chance of experiencing a major flood each year. Over the next 10 years, what is the probability that the district will experience:

(a) exactly no major floods?

(b) at most 3 major floods?

Solution:

This is a binomial distribution with:

n = 10 (number of trials, i.e., years)

p = 0.6 (probability of success, i.e., major flood)

q = 1 - p = 0.4 (probability of no major flood)

(a) $P(X = 0) = \binom{10}{0}(0.6)^0(0.4)^{10-0} = (0.4)^{10}$

(b) This means the district experiences 0, 1, 2, or 3 major floods:

$$P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$

Calculating each probability and summing them up, we get:

$$P(X \leq 3) \approx 0.0548$$

So, there is about 5.48% chance that the district will experience at most 3 major floods in the next 10 years.

Problem 5: A microfinance institution in Bangladesh has found that 15% of its borrowers cannot repay their loans on time. If the institution lends 20 new clients, what is the probability that,

a) twelve borrowers repay their loans on time

b) the expected number of clients who will repay their loans on time?

Solution:

a) $P(X = 12) = \binom{20}{12}(0.85)^{12}(0.15)^{20-12} = 0.0046$

The probability that exactly 12 borrowers will repay their loans on time is approximately 0.0046

b) In a binomial distribution, the expected value (mean) is given by:

$$E(X) = np$$

where:

n = 20

p = 0.85

In this case:

$$E(X) = 20 \times 0.85 = 17$$

Therefore, the expected number of clients who will repay their loans on time is 17.

Poisson distribution:

The Poisson distribution is widely used to model the probability of a given number of events occurring in a fixed interval of time or space when these events happen independently and at a constant average rate. In the context of Bangladesh, its applications are critical in areas such as healthcare, traffic management, and disaster preparedness. For instance, in public health, the Poisson distribution is used to analyze the occurrence of rare events like daily dengue fever cases in a specific region. It helps estimate the likelihood of a certain number of cases arising in a day, guiding resource allocation such as hospital beds or medical supplies. Similarly, in traffic management, the distribution can model the number of vehicles arriving at a traffic signal during a specific time interval, enabling authorities to optimize traffic flow. Moreover, the Poisson distribution is essential in disaster management, where it is used to model events such as floods, cyclones, or other natural calamities, helping to predict the frequency of these events and prepare accordingly. In business contexts, it is used to analyze customer arrival rates at service points, such as mobile financial services or ticket counters, helping organizations improve efficiency. Its versatility makes the Poisson distribution a valuable tool for analyzing and predicting event frequencies across diverse sectors in Bangladesh.

The probability distribution of a Poisson random variable is called a Poisson distribution, defined as:

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!};\ x = 0, 1, 2, \ldots;\ \lambda > 0.$$

Here, λ = The mean number of successes that occur in a specified region.

X = The actual number of successes that occur in a specified region.

The mean = variance = λ of the distribution. It is used, when certain conditions are met, as a probability distribution, where the possible number of discrete occurrences is much larger than the average number of occurrences in a given interval of time or space.

Poisson distribution vs Binomial distribution:

Poisson distribution is a limiting case of the binomial distribution under the terms: (a) $n \to \infty$ (very large), (b) $p \to 0$ (very small), and (c) $np = \lambda$; $\lambda > 0$. Poisson distribution has only one

parameter λ whereas binomial distribution has two parameters, *n* and *p*. Especially for very large values of *n* and very small values of *p* Binomial variable transform to Poisson variable.

Problem 1: In a particular neighborhood in Dhaka, power outages occur randomly at an average rate of 2 per month. What is the probability of:

(a) Exactly 3 power outages in a given month?

(b) No power outages in a given month?

Solution:

The probability mass function (PMF) of a Poisson distribution is:

$$\boldsymbol{P(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}}$$

Let, $X =$ number of power outages per month

$\lambda = 2$ (average number of power outages per month)

(a) $P(X = 3) = \frac{e^{-2}2^{3}}{3!} = 0.1804$

Therefore, the probability of exactly 3 power outages in a month is approximately 18.04%.

(b) $P(X = 0) = \frac{e^{-2}2^{0}}{0!} = 0.1353$

Therefore, the probability of no power outages in a month is approximately 13.53%.

Problem 2: On a particular stretch of a busy highway in Bangladesh, traffic accidents occur randomly at an average rate of 1.5 per day. What is the probability of:

(a) More than 2 accidents in a day?

(b) At most 1 accident in a day?

Solution:

Let, $X =$ number of accidents per day

$\lambda = 1.5$ (average number of accidents per day)

a) $P(X > 2) = 1 - P(x \leq 2) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$

$= 1 - \frac{e^{-1.5}1.5^{0}}{0!} - \frac{e^{-1.5}1.5^{1}}{1!} - \frac{e^{-1.5}1.5^{2}}{2!} = 0.1912$

Therefore, the probability of more than 2 accidents in a day is approximately 19.12%.

(b) $P(X \leq 1) = P(X = 0) + P(X = 1) = \frac{e^{-1.5}1.5^{0}}{0!} - \frac{e^{-1.5}1.5^{1}}{1!} = 0.5578$

Therefore, the probability of at most 1 accident in a day is approximately 55.78%.

Problem 3: A popular restaurant in Dhaka observes that customers arrive at an average rate of 3 per hour during peak hours. What is the probability that:

(a) Exactly 5 customers arrive in a particular hour?

(b) Fewer than 2 customers arrive in a particular hour?

Solution:

Let, $X =$ number of accidents per day

$\lambda = 3$ (average number of customers per hour)

(a) $P(X = 5) = \frac{e^{-3}3^5}{5!} = 0.1008$

So, the probability of exactly 5 customers arriving in an hour is approximately 10.08%.

(b) $P(X < 2) = P(X = 0) + P(X = 1) = \frac{e^{-3}3^0}{0!} + \frac{e^{-3}3^1}{1!} = 0.1991$

So, the probability of fewer than 2 customers arriving in an hour is approximately 19.91%.

Problem 4: In a certain region of Bangladesh, the average daily rainfall during the monsoon season is 5 millimeters. What is the probability that:

(a) There will be exactly 7 millimeters of rain on a given day?

(b) There will be more than 3 millimeters of rain on a given day?

Solution:

Let, $X =$ daily rainfall in millimeters

$\lambda = 5$ (average daily rainfall in millimeters)

(a) $P(X = 7) = \frac{e^{-5}5^7}{7!} = 0.1044$

So, the probability of exactly 7 millimeters of rain on a day is approximately 10.44%.

(b) $P(X > 3) = 1 - P(X \leq 3)$

$= 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)]$

$= 1 - (\frac{e^{-5}5^0}{0!} + \frac{e^{-5}5^1}{1!} + \frac{e^{-5}5^2}{2!} + \frac{e^{-5}5^3}{3!}) = 0.8753$

So, the probability of more than 3 millimeters of rain on a day is approximately 87.53%.

Problem 5: RFL assembles electric motors. The probability that a motor is defective is 0.01. What is the probability that a random sample of 200 motors will contain exactly 4 defective motors?

Solution: Let, $X =$ number of defective motors.

Given, $n = 200;\ p = 0.01$

So, $\lambda = np = 200 \times 0.01 = 2.$

Then,

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-2}2^x}{x!};\ x \geq 0.$$ Hence, $P(X = 4) = \frac{e^{-2}2^4}{4!} = 0.090.$

Geometric Distribution:

In a binomial distribution, the focus is on the number of successes in a fixed number of trials. However, when the interest shifts to determining the number of trials needed to observe the first success or the number of failures before the first success, the geometric probability distribution becomes relevant. This distribution models such uncertainty, where the probability of observing the first success after a certain number of trials (or failures) is calculated. In the context of Bangladesh, the geometric distribution is useful in various scenarios. For example, in public health studies, it can be applied to model how many vaccination attempts are needed to achieve the first successful immunization in rural areas where access to healthcare facilities might be inconsistent. Similarly, in agriculture, it can help predict how many planting attempts are required before achieving the first successful crop yield under uncertain weather conditions. Additionally, in business contexts like telecommunication, it can estimate how many customer calls are needed before closing the first sales deal. This probability model provides insights into the likelihood of success under uncertainty, enabling better decision-making and resource planning in real-world scenarios.

The probability function of the above-mentioned random variable is given

$$P\ (X = k) = p\ (1 - p)^{k-1}\ ,\ k = 1,2,3, \ldots \quad \text{or}$$

$$P\ (X = k) = p\ (1 - p)^{k}\ ,\ k = 0, 1,2, 3, \ldots$$

In both cases p is the probability of success, $(0 < p < 1)$.

Problem 1: A farmer in Bangladesh is searching for a suitable plot of land for cultivation. He inspects plots one by one, and each plot has a 20% chance of meeting his requirements. What is the probability that he finds a suitable plot on his:

(a) First attempt?

(b) Third attempt?

Solution:

p = 0.2 (probability of success, i.e., finding a suitable plot)

q = 1 - p = 0.8 (probability of failure)

(a) $P(k = 1) = 0.2\ (1\ -\ 0.2)^{1-1} = 0.2$

Therefore, the probability of finding a suitable plot on the first attempt is 20%.

(b) $P(k = 3) = 0.2\ (1\ -\ 0.2)^{3-1} = 0.128$

Therefore, the probability of finding a suitable plot on the third attempt is 12.8%.

Problem 2: A recent graduate is applying for jobs. He estimates that he has a 15% chance of getting an offer at each interview. What is the probability that he gets his first job offer on his fourth interview?

Solution:

p = 0.15 (probability of success, i.e., getting a job offer)

$q\ =\ 1\ -\ p\ =$ 0.85 (probability of failure)

$$P(k = 4) = 0.15\ (1\ -\ 0.15)^{4-1} = 0.0996$$

Therefore, the probability of getting his first job offer on the fourth interview is approximately 9.96%.

Problem 3: A quality control inspector in a Bangladeshi factory inspects products from an assembly line. The probability that a product is defective is 5%. What is the probability that the inspector finds the first defective product on the Tenth product inspected?

Solution:

$p\ =\ 0.05$ (probability of success, i.e., finding a defective product)

$q\ =\ 1\ -\ p\ =\ 0.95$ (probability of failure)

$$P(x = 10) = 0.05\ (1\ -\ 0.05)^{10-1} = 0.0315$$

So, the probability of finding the first defective product on the tenth product inspected is approximately 3.15%.

Problem 4: A marketing agency in Bangladesh conducts online advertising campaigns for its clients. They estimate that each campaign has a 30% chance of achieving its target goals. What is the probability that the agency achieves its target goals for the first time on Second campaign?

Solution:

p = 0.3 (probability of success, i.e., achieving target goals)

$q = 1 - p$ = 0.7 (probability of failure)

$$P(x = 2) = 0.3\ (1\ -\ 0.3)^{2-1} = 0.21$$

So, the probability of achieving target goals for the first time on the second campaign is 21%.

Example 5: A programmer in 70% cases becomes successful in developing program. Find the probability that in developing program at 8th attempt he / she fails.

Solution:

Let X be the number of times a programmer fails. Given $p = P$ [Failure] = 0.3.

$$P\ (x = 8) = p\ (1\ -\ p\ )^{8-1} = 0.3\ (0.7)^7 = 0.0247.$$

So, the probability of being successful at the eighth attempt is approximately 2.47%.

Exercises

Binomial Distribution

1. For a binomial distribution, mean is 2 and variance is 1. Estimate n, p and q.
2. A survey in a Dhaka slum finds that 70% of households have access to clean drinking water. If 10 households are randomly selected, what is the probability that at best 7 households have access to clean drinking water?
3. A farmer in a rural area of Bangladesh plants 15 mango trees. Based on experience, each tree has a 60% chance of producing fruit in a given season. What is the probability that at best 9 mango trees will produce fruit in the next season?
4. A study in a Bangladeshi village shows that 80% of children aged 5-11 are enrolled in primary school. If 12 children from this age group are randomly selected, what is the probability that a) exactly 10 of them are enrolled in primary school? b) none of them are enrolled in primary school

5. A microfinance institution in Bangladesh observes that 95% of its loans to small businesses for poultry farming are repaid on time. If the institution provides loans to 20 small businesses, what is the probability that exactly 18 of them will repay their loans on time?
6. A new COVID-19 vaccine is being tested in Bangladesh. Clinical trials show that the vaccine is 80% effective in preventing symptomatic infection. If 1000 people in Dhaka are vaccinated, what is the probability that, (a) Exactly 750 people will be protected from symptomatic infection? (b) At least 850 people will be protected?
7. Apollo hospital records show that 75% of patients suffering from kidney disease die of it. What is the probability that out of 6 randomly selected patients, 4 will recover?
8. A textile factory in Bangladesh produces bolts of cloth. Historical data shows that 20% of bolts have minor defects. If a random sample of 20 bolts is inspected, what is the probability that exactly 1 bolt will have a minor defect?
9. A survey in a Bangladeshi university finds that 85% of students own a smartphone. If a random sample of 20 students is selected, what is the probability that fewer than 15 students own a smartphone?
10. 60% of the streetlights in Dhaka city work properly. One day 25 streetlights were selected randomly. What was the expected number of streetlights working properly along its variance?

Poisson Distribution

1. A busy intersection in Dhaka experiences an average of 2 traffic accidents per week. What is the probability that there will be at least 3 accidents next week?
2. A popular restaurant in Dhaka receives an average of 5 customer complaints per day. What is the probability that the restaurant will receive no complaints on a given day?
3. A textile factory in Bangladesh observes an average of 3 machine breakdowns per month. What is the probability that there will be at most 2 machine breakdowns next month?
4. A call center in Dhaka receives an average of 20 calls per hour. What is the probability that the call center will receive fewer than 15 calls in the next hour?
5. Suppose 300 misprints are distributed randomly throughout a book of 500 pages. Find the probability that a given page contains: Exactly 4 misprints, (b) 2 or more misprints
6. Two percent handsets produced by Walton are usually found defective. Walton produces 300 sets per day. Find the probability that there will be (a) 4 defective sets (b) at best 3 defective sets (c) 2 to 5 defective sets in a day's production.

7. Let *X* be the number of typos on a printed page with a mean of 4 typos per page. What is the probability that a randomly selected page has a) at least two typo b) at best 3 typo c) exactly 5 typos on it?

Geometric Distribution

1. A farmer in Bangladesh inspects mango trees one by one to identify those infested with pests. If 20% of the trees are infested, what is the probability that the farmer finds the first infested tree on the third tree inspected?
2. A student applies to different universities in Bangladesh. If the probability of getting accepted to any given university is 30%, what is the probability that the student gets accepted to the first university on their fourth application?
3. In a class 40% are female students. Find the probability that the 9th student enter in the class will be a male.
4. A telemarketer makes cold calls to potential customers in Bangladesh. If the probability of making a successful sale on any given call is 10%, what is the probability that the telemarketer makes their first sale on the fifth call?
5. A fisherman in Bangladesh casts his net repeatedly in a lake. If the probability of catching a fish on any given cast is 25%, what is the probability that the fisherman catches his first fish on the third cast?